

VIT AP DSA JAVA Lab — All Questions with Answers (Compiled & Run)

All 15 questions were typed manually, compiled with `javac`, and executed with `java`. Below are the complete results.

Question 1 (Q1 / 15) — Row-wise Minimum Product

Java

```
class Test {
    public static void main(String[] args) {
        int[][] a = { {2,4,6}, {8,10,12}, {14,16,18} };
        int result = 1;
        for (int i = 0; i < a.length; i++) {
            int min = a[i][0];
            for (int j = 1; j < a[i].length; j++)
                min = Math.min(min, a[i][j]);
            result *= min;
        }
        System.out.println(result);
    }
}
```

Output: 224

Explanation: Row mins are 2, 8, 14 → product = $2 \times 8 \times 14 = 224$.

Question 2 (Q3 / 15) — Modulo with Negated Array Elements

Java

```
public class Main {
    public static void main(String[] args) {
        int[] arr = {7, 9, 1, 3, 5};
        int result = 0;
        for (int i = 0; i < arr.length; i++) {
            result %= -arr[i];
        }
        System.out.println(+result);
    }
}
```

```
}  
}
```

Output: 0

Explanation: $0 \% \text{ anything}$ is always 0, so result stays 0 throughout.

Question 3 (Q4 / 15) — Product of Transformed Array

Java

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {5, 2, 3, 2};  
        int[] result = new int[arr.length];  
        for (int i = 0; i < arr.length; i++) {  
            result[i] = (arr[i] + arr[i] / 5) * 2;  
        }  
        int product = 1;  
        for (int i = 0; i < result.length; i++) {  
            product *= result[i];  
        }  
        System.out.println(product);  
    }  
}
```

Output: 1152

Explanation: $\text{result} = \{12, 4, 6, 4\} \rightarrow \text{product} = 12 \times 4 \times 6 \times 4 = 1152$.

Question 4 (Q5 / 15) — Successive Modulo on Array

Java

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {4, 8, 12, 16};  
        int result = 0;  
        for (int i = 0; i < arr.length; i++) {  
            result %= arr[i];  
        }  
        System.out.print(result);  
    }  
}
```

```
}  
}
```

Output: 0

Explanation: $0 \% 4 = 0$, then $0 \% 8 = 0$, etc. Result remains 0.

Question 5 (Q6 / 15) — Sum of $(arr[i] - 3) + (arr[i] / 3)$

Java

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {6, 9, 12, 15};  
        int[] result = new int[arr.length];  
        for (int i = 0; i < arr.length; i++) {  
            result[i] = (arr[i] - 3) + (arr[i] / 3);  
        }  
        int sum = 0;  
        for (int i = 0; i < result.length; i++) {  
            sum += result[i];  
        }  
        System.out.print(sum);  
    }  
}
```

Output: 44

Explanation: $result = \{5, 9, 13, 17\} \rightarrow sum = 5 + 9 + 13 + 17 = 44$.

Question 6 (Q7 / 15) — Matrix Multiplication $b[0][1]$

Java

```
class Test {  
    public static void main(String[] args) {  
        int[][] a = { {1,2}, {3,4} };  
        int[][] b = new int[2][2];  
        for (int i = 0; i < 2; i++)  
            for (int j = 0; j < 2; j++)  
                for (int k = 0; k < 2; k++)  
                    b[i][j] += a[i][k] * a[k][j];  
        System.out.println(b[0][1]);  
    }  
}
```

```
}  
}
```

Output: 10

Explanation: $b[0][1] = a[0][0] * a[0][1] + a[0][1] * a[1][1] = 1 * 2 + 2 * 4 = 2 + 8 = 10$.

Question 7 (Q8 / 15) — Column-wise Maximum of Matrix

Java

```
class Main {  
    public static void main(String[] args) {  
        int[][] matrix = { {3,7,2}, {5,1,8}, {9,6,4} };  
        for (int j = 0; j < matrix[0].length; j++) {  
            int max = matrix[0][j];  
            for (int i = 1; i < matrix.length; i++) {  
                if (matrix[i][j] > max)  
                    max = matrix[i][j];  
            }  
            System.out.print(max + " ");  
        }  
    }  
}
```

Output: 9 7 8

Explanation: Col 0 max = max(3,5,9) = 9; Col 1 max = max(7,1,6) = 7; Col 2 max = max(2,8,4) = 8.

Question 8 (Q9 / 15) — Zigzag Column Traversal Sum

Java

```
class Test {  
    public static void main(String[] args) {  
        int[][] a = { {1,2,3}, {4,5,6}, {7,8,9} };  
        int sum = 0;  
        for (int j = 0; j < 3; j++) {  
            if (j % 2 == 0) {  
                for (int i = 0; i < 3; i++) sum += a[i][j];  
            } else {  
                for (int i = 2; i >= 0; i--) sum += a[i][j];  
            }  
        }  
    }  
}
```

```

    }
    System.out.println(sum);
}
}

```

Output: 45

Explanation: Col 0 (down): $1+4+7 = 12$; Col 1 (up): $8+5+2 = 15$; Col 2 (down): $3+6+9 = 18 \rightarrow$ total = 45.

Question 9 (Q10 / 15) — Count Negative Even Numbers

Java

```

class Test {
    public static void main(String[] args) {
        int[][] a = { {1, -2, 1}, {-4, 5, -6}, {7, -8, 9} };
        int count = 0;
        for (int[] row : a)
            for (int x : row)
                if (x < 0 && x % 2 == 0)
                    count++;
        System.out.println(count);
    }
}

```

Output: 4

Explanation: Negative even numbers: -2, -4, -6, -8 $\rightarrow$ count = 4.

Question 10 (Q11 / 15) — Most Frequent Element (First to Reach Max)

Java

```

class Main {
    public static void main(String[] args) {
        int[] arr = {1, 3, 2, 1, 4, 3};
        int maxCount = 0, element = -1;
        for (int i = 0; i < arr.length; i++) {
            int count = 0;
            for (int j = 0; j < arr.length; j++)
                if (arr[j] == arr[i]) count++;

```

```

        if (count > maxCount) {
            maxCount = count;
            element = arr[i];
        }
    }
    System.out.println(element);
}
}

```

Output: 1

Explanation: Both 1 and 3 appear twice, but `>` (not `>=`) means the first to reach count=2 wins. Element 1 at index 0 reaches count=2 first → 1.

Question 11 (Q12 / 15) — Sum of `-(arr[i] * 2)`

Java

```

public class Main {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5};
        int[] result = new int[arr.length];
        for (int i = 0; i < arr.length; i++) {
            result[i] = (arr[i] - arr[i]) - (arr[i] * 2);
        }
        int sum = 0;
        for (int i = 0; i < result.length; i++) {
            sum += result[i];
        }
        System.out.println(sum);
    }
}

```

Output: -30

Explanation: Each `result[i] = 0 - arr[i]*2` → -2, -4, -6, -8, -10 → sum = -30.

Question 12 (Q13 / 15) — Row Index with Most 1s

Java

```

class Main {
    public static void main(String[] args) {
        int[][] arr = {

```

```

        {0, 1, 1, 1},
        {0, 0, 1, 1},
        {1, 1, 1, 1},
        {0, 0, 0, 0}
    };
    int maxCount = 0, rowIndex = -1;
    for (int i = 0; i < arr.length; i++) {
        int count = 0;
        for (int j = 0; j < arr[0].length; j++) {
            if (arr[i][j] == 1) count++;
        }
        if (count > maxCount) {
            maxCount = count;
            rowIndex = i;
        }
    }
    System.out.println(rowIndex);
}

```

Output: 2

Explanation: Row counts of 1s → Row 0: 3, Row 1: 2, Row 2: 4, Row 3: 0 → rowIndex = 2.

Question 13 (Q14 / 15) — Sum of Both Diagonals of 4×4 Matrix

Java

```

class Test {
    public static void main(String[] args) {
        int[][] a = {
            {1, 2, 3, 4},
            {5, 6, 7, 8},
            {9, 10, 11, 12},
            {13, 14, 15, 16}
        };
        int sum = 0;
        for (int i = 0; i < 4; i++) {
            sum += a[i][i];
            sum += a[i][3 - i];
        }
        System.out.println(sum);
    }
}

```

```
}  
}
```

Output: 68

Explanation: Main diagonal = $1+6+11+16 = 34$; Anti-diagonal = $4+7+10+13 = 34$; total = **68**.
(Note: center elements 6 and 11 each counted twice, which is intended by the code.)

Question 14 (Q15 / 15) — Sum of $(arr[i]*arr[i] + 5) / 2$

Java

```
public class Main {  
    public static void main(String[] args) {  
        int[] arr = {2, 4, 6, 8, 10};  
        int[] result = new int[arr.length];  
        for (int i = 0; i < arr.length; i++) {  
            result[i] = (arr[i] * arr[i] + 5) / 2;  
        }  
        int sum = 0;  
        for (int i = 0; i < result.length; i++) {  
            sum += result[i];  
        }  
        System.out.println(sum);  
    }  
}
```

Output: 120

Explanation: $result = \{4, 10, 20, 34, 52\} \rightarrow sum = 4 + 10 + 20 + 34 + 52 = 120$.

Question 15 (Q12 from second section) — Reverse Loop with Break

Java

```
public class Main {  
    public static void main(String[] args) {  
        int[] numbers = {5, 10, 15, 20, 25};  
        int result = 0;  
        String flag = "running";  
        for (int i = numbers.length - 1; i >= 0; i--) {  
            result += numbers[i];  
        }  
    }  
}
```

```

        if (result > 30) {
            flag = "stopped";
            break;
        }
    }
    System.out.println(result);
}
}

```

Output: 45

Explanation: Reverse order: 25 → result=25 (≤ 30), then 20 → result=45 (> 30 , break) → **45**.

Summary Table

Question No.	Description	Output
Q1	Row-wise min product	224
Q3	result $\% = -arr[i]$	0
Q4	Product of transformed array	1152
Q5	Successive modulo from 0	0
Q6	Sum of $(arr[i]-3)+(arr[i]/3)$	44
Q7	Matrix multiplication $b[0][1]$	10
Q8	Column-wise max of matrix	9 7 8
Q9	Zigzag column traversal sum	45
Q10	Count negative even numbers	4
Q11	Most frequent element (first wins)	1
Q12	Sum of $-(arr[i]*2)$	-30
Q13	Row index with most 1s	2
Q14	Both diagonals sum of 4×4	68
Q15	Sum of $(arr^2+5)/2$	120
Q12b	Reverse loop with break > 30	45